

User & Administrator Guide: Dynamic Command Configuration Engine (`axi_command_config`)

Document Version: 1.0.0
Target Audience: Axipt Developers, System Administrators, Technical QA Engineers
Applies To: Axi Command Palette (`Axi_Beta`) v0.4.0+ / Axipt Web 11.4+

1. Overview & Purpose

The **Dynamic Command Configuration Engine** is a metadata-driven navigation architecture in the Axi Command Palette. Historically, adding or modifying command navigation routes required editing JavaScript source code (`axicmdmain.js`).

With the introduction of the `axi_command_config` table and REST API, **100% of command navigation is managed directly in the database**. System administrators and developers can register new commands, link them to forms, reports, or custom pages, and configure contextual parameters **without changing a single line of JavaScript**.

2. Database Schema Reference (`axi_command_config`)

The table `axi_command_config` is stored in the application database (PostgreSQL or Oracle).

Column Specifications

Column Name	Data Type	Nullable	Description & Purpose
<code>config_id</code>	<code>VARCHAR(50)</code>	No	Primary Key. Unique identifier for the configuration rule (e.g. <code>cfg_configure_user</code>).
<code>command</code>	<code>VARCHAR(50)</code>	No	The root command verb (e.g. <code>Configure</code> , <code>SDK</code> , <code>Upload</code> , <code>Download</code> , <code>View</code> , <code>Create</code> , <code>Edit</code>).
<code>prompt_options</code>	<code>VARCHAR(200)</code>	No	The specific prompt option value to match (e.g. <code>user</code> , <code>role listing</code> , <code>app</code> , <code>default</code>). Case-insensitive.
<code>prompt_id</code>	<code>VARCHAR(50)</code>	No	Target identifier: <code>TransID</code> for Tstructs, <code>IViewName</code> for Iviews, or filename for

Column Name	Data Type	Nullable	Description & Purpose
			custom pages.
prompt_option_type	VARCHAR(20)	No	Execution mode determining how the client shell handles the navigation.
param_field	VARCHAR(100)	Yes	Query parameter key to bind the user's input parameter to (e.g. name , type , actorname).
target_url	VARCHAR(500)	Yes	Explicit target URL template override. If omitted, constructed dynamically based on prompt_option_type .
extra_params	VARCHAR(500)	Yes	Additional query string parameters or dynamic tokens to append (e.g. &status=true&user=:username).
active	VARCHAR(1)	Yes	Status flag ('T' for Active, 'F' for Disabled). Default: 'T' .

3. Supported Navigation Modes (prompt_option_type)

The engine provides specialized execution handlers tailored to Axpert structure types:

1. tstruct (Transaction Forms)

- **Use Case:** Opens an Axpert Tstruct form for creating a new record or editing an existing record.
- **Behavior:** If user provides a search value and param_field is specified, it opens in edit mode (hltype=load); otherwise opens in create mode (hltype=open).
- **Example:**

```
INSERT INTO axi_command_config (config_id, command, prompt_options,
prompt_id, prompt_option_type, param_field, active)
VALUES ('cfg_configure_user', 'Configure', 'user', 'axusr', 'tstruct',
'uname', 'T');
```

2. iview (Interactive Views / Reports)

- **Use Case:** Opens an Axpert interactive view report.
- **Behavior:** Routes directly to iview.aspx?ivname=<prompt_id> .
- **Example:**

```
INSERT INTO axi_command_config (config_id, command, prompt_options,
prompt_id, prompt_option_type, active)
VALUES ('cfg_configure_users', 'Configure', 'user listing', 'axusers',
'iview', 'T');
```

3. tstruct/iview or iview/tstruct (Dual-Mode Conditional Routing)

- **Use Case:** Single command that opens a listing report by default, but directly opens a specific record form if an entity name/ID is supplied.
- **Syntax for prompt_id:** <TstructID>/<IviewName> (e.g. ad_ur/ad___url).
- **Behavior:**
 - If user types `Configure Role` → Opens IView report `ad___url` .
 - If user types `Configure Role "Admin"` → Opens Tstruct form `ad_ur` with record `Admin` .

- **Example:**

```
INSERT INTO axi_command_config (config_id, command, prompt_options,
prompt_id, prompt_option_type, param_field, active)
VALUES ('cfg_role_dual', 'Configure', 'role', 'ad_ur/ad___url',
'tstruct/iview', 'rname', 'T');
```

4. url (Standard Custom ASPX / HTML Pages)

- **Use Case:** Redirects to custom Axpert web pages, tools, or configuration dashboards.
- **Behavior:** Resolves relative URLs to `../aspx/<prompt_id>` or uses explicit `target_url` .
- **Example:**

```
INSERT INTO axi_command_config (config_id, command, prompt_options,
prompt_id, prompt_option_type, param_field, target_url, active)
VALUES ('cfg_config_app', 'Configure', 'app', 'appconfig.aspx', 'url',
NULL, '../aspx/appconfig.aspx', 'T');
```

5. processflow or url/tstruct (Process Builder Flow)

- **Use Case:** Opens visual workflow builder diagrams.
- **Behavior:** Routes to `processflow.aspx?loadcaption=AxProcessBuilder&processname=<paramValue>` . If no parameter is passed, falls back to `ad_pm` Tstruct.
- **Example:**

```
INSERT INTO axi_command_config (config_id, command, prompt_options,
prompt_id, prompt_option_type, param_field, target_url, active)
```

```
VALUES ('cfg_configure_peg', 'Configure', 'peg', 'ad_pm', 'processflow',
'processname', '../aspx/processflow.aspx?loadcaption=AxProcessBuilder',
'T');
```

6. ivtoivload (IView-to-IView Parameterized Navigation)

- **Use Case:** Navigates to parameterized summary Iviews with preset filter bindings.
- **Example:**

```
INSERT INTO axi_command_config (config_id, command, prompt_options,
prompt_id, prompt_option_type, param_field, target_url, extra_params,
active)
VALUES ('cfg_configure_dim_list', 'Configure', 'dimension listing',
'ad___upg', 'ivtoivload', 'prole', '../aspx/ivtoivload.aspx?
ivname=ad___upg', 'AxOpenAct=true&isDupTab=false', 'T');
```

4. Dynamic URL Placeholders

You can include contextual placeholders in `target_url` and `extra_params`. The client engine resolves these dynamically from the logged-in session before launching the target:

Placeholder Token	Resolved Runtime Value	Description
:username	window.mainUserName	The login username of the current user.
:userroles	window.AxUserRoles	Comma-separated list of active user roles.
:userresp	window.userResp	Active responsibility of the logged-in user.
:appname	appname / window.mainProject	Current application project schema name.
:param	paramValue	The parameter value typed or selected by the user.
:paramField	paramField	The field name configured for the target.

Placeholder Usage Example

```
target_url: "../aspx/customdashboard.aspx?"
```

```
user=:username&role=:userroles&proj=:appname"
```

✂ 5. Step-by-Step Tutorial: Configuring a New Command

Scenario

You want to add a new command: **Configure SLA "Gold Policy"** that opens custom page `SlaConfig.aspx` with the selected policy name and current user's role.

Step 1: Register Autocomplete Prompt

If your new option is under **Configure** (`cmdtoken = 4`) or **SDK** (`cmdtoken = 7`), ensure it is registered in `axi_command_prompts` so users receive auto-complete suggestions:

Option A: For **Configure** Commands (`cmdtoken = 4`)

```
-- 1. Append sub-option name to Position 2 (Object Type prompt values)
UPDATE axi_command_prompts
SET promptvalues = promptvalues || ',SLA'
WHERE cmdtoken = 4 AND wordpos = 2;

-- 2. Append matching data source to Position 3 (Object Name data source)
-- Use 'Axi_Dummy' if the sub-option is static (no 3rd token suggestion
required)
UPDATE axi_command_prompts
SET promptsource = promptsource || ',Axi_Dummy'
WHERE cmdtoken = 4 AND wordpos = 3;

-- OR if dynamic: specify your custom Axpert Data Source (e.g.,
',axi_custom_ds')
-- UPDATE axi_command_prompts SET promptsource = promptsource ||
',axi_custom_ds' WHERE cmdtoken = 4 AND wordpos = 3;
```

Option B: For **SDK** Commands (`cmdtoken = 7`)

```
-- 1. Append sub-option name to Position 2 (SDK Type prompt values)
UPDATE axi_command_prompts
SET promptvalues = promptvalues || ',My Custom Tool'
WHERE cmdtoken = 7 AND wordpos = 2;

-- 2. Append matching data source to Position 3 (SDK Name data source)
-- Use 'Axi_Dummy' if static
```

```

UPDATE axi_command_prompts
SET promptsource = promptsource || ',Axi_Dummy'
WHERE cmdtoken = 7 AND wordpos = 3;

-- OR if dynamic: specify your custom Axi Data Source (e.g.,
',axi_custom_ds')
-- UPDATE axi_command_prompts SET promptsource = promptsource ||
',axi_custom_ds' WHERE cmdtoken = 7 AND wordpos = 3;

```

Tip

Dynamic Suggestions via axdirectsql (Custom ADS):

If your sub-option requires dynamic auto-complete suggestions for Position 3 (e.g., listing all SLA policies or custom tools from database):

1. Create a data source entry in the `axdirectsql` table (e.g., `axi_custom_ds`) returning name and `displaydata`.
2. Append that data source name into Position 3 (`wordpos = 3`) of `axi_command_prompts` as `promptsource = promptsource || ',axi_custom_ds'` instead of `Axi_Dummy`.

Step 2: Insert the Configuration Row

Execute the following SQL statement in your database:

```

INSERT INTO axi_command_config (
    config_id,
    command,
    prompt_options,
    prompt_id,
    prompt_option_type,
    param_field,
    target_url,
    extra_params,
    active
) VALUES (
    'cfg_configure_sla',
    'Configure',
    'sla',
    'SlaConfig.aspx',
    'url',
    'policyname',
    '../aspx/SlaConfig.aspx',

```

```
'role=:userroles&user=:username',  
'T'  
);
```

Step 3: Test in the Command Palette

1. Open Axipt Web Shell and press `Ctrl + Space` to activate Axi Command Palette.
2. Type: `Configure SLA "Gold Policy"`
3. Press `Enter` or click **Go**.
4. The system executes:

```
../aspx/SlaConfig.aspx?polycname=Gold%20Policy&role=Developer&user=admin
```

5. Click **Pop** to test tabbed execution in `PopupContainer.html`.

6. Troubleshooting & FAQs

Q1: I added a row in `axi_command_config`, but the command palette says "Unknown command".

- **Check Cache:** The configuration is loaded when the shell initializes. Refresh the browser (`Ctrl + F5`) or click the **Refresh** button on the Axi toolbar.
- **Check Matching:** Verify that `command` matches the root verb exactly (e.g. `Configure`) and `prompt_options` matches the typed sub-verb in lowercase.

Q2: How do I create a fallback route for an entire command group?

- Set `prompt_options` to `'default'`. If no exact sub-option matches, the engine falls back to this rule:

```
INSERT INTO axi_command_config (config_id, command, prompt_options,  
prompt_id, prompt_option_type, active)  
VALUES ('cfg_default_sdk', 'SDK', 'default', 'devexplorer', 'url', 'T');
```

Q3: My custom page opens in the main window, but I want it to open in a popup tab.

- Users can press `Ctrl + Shift + Enter` or select the **Pop** action button. The engine automatically appends `&AxiPop=true` and loads the page within `PopupContainer.html`.

Q4: Can I temporarily disable a command without deleting the row?

- Yes. Update the status flag: `UPDATE axi_command_config SET active = 'F' WHERE config_id = 'cfg_configure_sla';`